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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-46

Transformer Design-III

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Recap

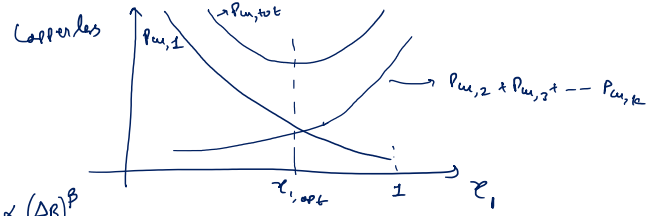
- We have studied
  - Core loss

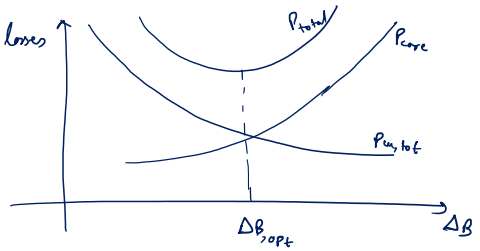
$$P_{core} = K_{core}(\Delta B)^{\beta} A_c l_c$$

  - Maximum flux density

$$\Delta B = \frac{\lambda_1}{2 n_1 A_c}$$

  - Winding losses
  - Optimum value of ΔB





Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Summary

Specified quantities

Wire resistivity → $\rho\ (\Omega\text{-cm})$ (copper wire $\rho = 1.724 \times 10^{-6}\ \Omega\text{-cm}$ at 25°C)

Total rms winding current, referred to winding 1, $I_{tot} = \sum_{j=1}^{k} \left(\frac{n_j}{n_1}\right) \cdot I_j$

Desired turns ratio → $\frac{n_2}{n_1}, \frac{n_3}{n_1}, \cdots \frac{n_k}{n_1}$

Applied volt-sec of winding 1, $A_1 = \int_{\text{positive portion of } v_1(t)} v_1(t)\,dt$ (volt-sec)

Allowed total power loss, $P_{total,allow}$ (W)

winding fill factor, K_u

Core loss exponent, β (2.5 – 2.7)

Core loss coefficient, K_{core}

Design outcome

Core cross-sectional Area, A_c
Core window Area, W_a
Mean length per turn, MLT
Magnetic path length, l_c
} Core dimensions

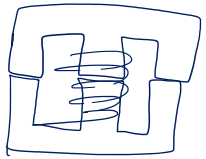
Peak ac flux density, ΔB (T)

Wire size (cross sectional area of wire), $A_{w,1}\ A_{w,2} \cdots A_{w,k}$

no. of turns

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Transformer Design Procedure



Step-1

Select the core

EE → EE5, EE25, EE30 - -

↳ A_c, l_c, MLT, w_a / β, K_{core} ~ core material

Step-2

Evaluate, $P_{core} = K_{core} (\Delta B)^{\beta} A_c l_c$ in terms of ΔB

Evaluate, $P_{w,tot} = \frac{8(MLT)}{w_a K_u} \times \frac{d_1^2}{4(\Delta B)^2} A_c \sum_{j=1}^{K} \left(\frac{n_j}{n_1}\right)^2 \frac{I_j^2}{\rho_j^2}$ in term of ΔB → $\rho_{j,opt}$

optimum value of ρ_j (i.e, $\rho_{j,opt}$)

Such that $P_{w,tot}$ is the minimum value

Step-3

we will obtain the optimum of ΔB

⇒ By $\frac{d(P_{total})}{d(\Delta B)} = 0$ (where $P_{total} = P_{core} + P_{w,tot}$)

⇒ $\frac{d(P_{core})}{d(\Delta B)} = -\frac{d(P_{w,tot})}{d(\Delta B)}$ ⇒ $\Delta B_{,opt}$

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Transformer Design Procedure Cont.

Step-4

Evaluate $P_{total} = P_{cu,tot}\big|_{\Delta B_{opt}} + P_{core}\big|_{\Delta B_{opt}}$

such that $P_{total} \leq P_{total,allow}$ → specified quantities

→ defines whether the next iteration is to be done or not

Step-5

the number of turns in the winding

$$n_1 = \frac{d_1}{2 \Delta B A_c}$$

d_1 → specified quantity

A_c → the value from selected core size

ΔB → ΔB_{opt}

⇒ the $n_2 = \left(\frac{n_2}{n_1}\right) n_1$

$\left(\frac{n_2}{n_1}\right)$ → the specified

$n_3 = \left(\frac{n_3}{n_1}\right) n_1$

$\vdots$

$n_k = \left(\frac{n_{1c}}{n_1}\right) n_1$

→ $n_1, n_2, n_3 \cdots n_k$

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Transformer Design Procedure Cont.

Step-6

⇒ Then obtain the Area of cross section of wire

$$A_{w,1} \le \frac{r_{1,opt} K_u W_a}{n_1}$$

$r_{1,opt}$ → obtain from the selected core

K_u → winding fill factor

n_1 → Step-4

Step-2

$$A_{w,2} \le \frac{r_{2,opt} K_u W_a}{n_2}$$

$$\vdots$$

$$A_{w,K} \le \frac{r_{K,opt} K_u W_a}{n_K}$$

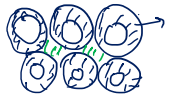
$A_{w,1}$; $A_{w,2}$; $A_{w,3}$ → appropriate wire gauge

AWG or SWG

AWG8, AWG10 - - - - AWG40, AWG44 - -



$K_u < 1$



→ enamel coating

⇒ Assume, the wire geometry, skin and proximity effect are not there

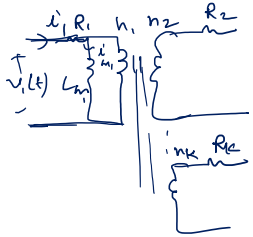
$$I^2 \frac{\rho l}{A}$$

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Transformer Design Procedure Cont.

- Core dimensions → A_c, l_c, w_a, MLT
- Peak ac flux density → $\Delta B_{,opt}$ → for which the P_{total} is minimum
- $f_{s,opt}$ → $P_{w,tot}$ is minimum
- $A_{w,1}$; $A_{w,2}$; $A_{w,3}$; - - - $A_{w,k}$
- n_1, n_2, n_3 - - - n_k

Transformer Design Check



Magnetizing inductance referred to primary

$$L_{m1} = \frac{\mu_c n_1^2 A_c}{l_c}$$

Peak ac magnetizing current, $i_{m,pk} = \frac{d_1}{2 L_{m1}}$

DC winding resistance:

$$R_1 = \frac{\rho n_1 (MLT)}{A_{w,1}}$$

$$\vdots$$

$$R_K = \frac{\rho n_K (MLT)}{A_{w,K}}$$

Evaluate losses (core as well as copper), and then check if it is within the allowed limit of P_{tot}

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Thank You